

# Real-Time Context-Aware Sign Language to Multilingual Speech Conversion System with Emotion Detection

## Problem Statement

Communication barriers between deaf and hard-of-hearing individuals and the hearing population represent a significant global challenge. Over 430 million people worldwide require rehabilitation for disabling hearing loss, projected to exceed 700 million by 2050 (World Health Organization, 2023), costing the global economy approximately US\$980 billion annually through lost productivity, healthcare expenses, and educational inequality (WHO, 2021). Despite this scale, existing sign language recognition systems remain critically limited. Most deployed systems perform direct word-by-word translation without grammatical restructuring, producing unnatural and incomprehensible spoken output, a fundamental problem given that sign languages possess entirely distinct grammatical structures from spoken languages (Khan et al., 2024). Furthermore, virtually no system addresses low-resource languages such as Nepali Sign Language, leaving deaf communities across South Asia severely underserved, as labeled datasets and speech-output systems for NSL remain almost entirely absent (arXiv, 2025). Finally, emotional awareness is entirely overlooked; facial expressions carry critical grammatical and emotional meaning in sign language (Papastratis et al., 2024), yet current multimodal systems have not adequately integrated emotion recognition into sign-to-speech pipelines (Lian et al., 2023), meaning tone, urgency, and feeling are lost in translation.

This project addresses these three compounding gaps, grammatical inadequacy, linguistic exclusion of Nepali, and absence of emotion-aware speech modulation, by developing a single integrated, real-time AI system that converts sign language gestures into grammatically correct, emotionally modulated speech in both English and Nepali.

## References:

1. World Health Organization (2023) *Deafness and hearing loss*.
2. World Health Organization (2021) *World Report on Hearing*. Geneva: WHO.
3. Papastratis, I. et al. (2024) 'Reviewing 25 years of continuous sign language recognition research', *Information Processing & Management*, 61(5).
4. Khan, N.S. et al. (2024) 'Enhancing sign language recognition using CNN and SIFT', *Journal of King Saud University – Computer and Information Sciences*, 36(2).
5. Sunuwar, J. et al. (2025) *Nepali Sign Language Characters Recognition: Dataset Development and Deep Learning Approaches*. arXiv:2510.11243.
6. Lian, H. et al. (2023) 'A survey of deep learning-based multimodal emotion recognition', *Entropy*, 25(10), p.1440.
7. Scientific Reports (2026) 'Smart comprehend gesture-based emotions recognition system for people with hearing disability'. *Nature/Scientific Reports*.

## **Research Questions**

RQ1. To what degree does integrating context-aware NLP correction into a CNN-LSTM-based sign language recognition pipeline improve the grammatical fluency of synthesized English and Nepali speech output compared to baseline word-by-word translation?

RQ2. How do deaf and hard-of-hearing users in Nepal perceive the usability and communicative effectiveness of an AI-driven sign language to multilingual speech system during real-world interaction tasks?

RQ3. How does the inclusion of a Transformer-based NLP correction module affect the grammatical fluency score of sign-to-speech output when compared to a system without contextual correction?

## **Research Aim**

This project aims to design and develop a real-time, context-aware sign language to multilingual speech conversion system that addresses the grammatical, linguistic, and emotional gaps in current assistive communication technology, with specific focus on English and Nepali language support for deaf and hard-of-hearing individuals.

## **SMART Objective**

1. **Sign Recognition:** To develop a deep learning-based gesture recognition model using CNN and LSTM architectures capable of recognizing a minimum of 50 static and dynamic signs with at least 85% classification accuracy.
2. **NLP Grammar Correction:** To implement a context-aware NLP module using Transformer-based models that restructures raw sign sequences into grammatically correct English sentences, achieving at least 20% improvement in fluency score over baseline word-by-word output.
3. **Multilingual Speech Synthesis:** To integrate a multilingual text-to-speech engine supporting both English and Nepali language output with a system response latency of under 1.5 seconds per sentence.
4. **Emotion Detection:** To design and deploy a facial expression recognition module capable of classifying at least four emotions — happy, sad, angry, and neutral — with a minimum accuracy of 80%, and use the output to modulate TTS speech parameters (pitch, rate) accordingly.

5. Evaluation: To evaluate overall system performance using quantitative metrics including accuracy, F1-score, and latency, and conduct usability testing with a minimum of 10 participants from the deaf or hard-of-hearing community, achieving a user satisfaction score of at least 80%.